ST-HGRec — Technical Gist

Trust-aware sentiment-graph recommendation with faithful aspect-level explanations

Offline pipeline summary — 2026-09-07

One paragraph. We recommend Amazon *Video Games* items with a heterogeneous LightGCN (**ST-HGRec**) that runs over a graph of *user / item / aspect* nodes. Aspect-based sentiment (PyABSA) and per-user trust signals enter the graph as edge structure and edge weights — never as an MLP. The model exposes hooks that let an *honest* explanation cite only graph facts; an *ungrounded* generator (study stimulus, quarantined) cites nothing real; a remove-a-reason *faithfulness* check separates the two. The paper's contribution is a human calibration study — the offline system only has to be **solid and reproducible, not best-in-class**.

1 Data and split

Amazon Reviews 2023, Video Games category (McAuley-Lab). We use the **official 5-core timestamp benchmark split verbatim** — RecBole loads `video_games.{train,valid,test}.inter` through its `benchmark_filename` mechanism, which partitions by file and *bypasses* any random re-split.

	train	valid	test
interactions	736,827	34,510	43,249

Atomic-file row counts equal the official CSVs exactly (±0). **Leakage guard:** the graph, trust signals, and ABSA population are all built from `train.inter` *only*. Train population: 91,562 users, 22,976 items, ≈748K reviews with usable text.

2 Pipeline

`raw parquet` → `build_atomic (5-core split → .inter/.item/.user)`
→ `[ run_baselines ]  [ trust_signals ]  [ run_absa → normalize_aspects ]`
→ `graph_tensors (per-relation sparse` $\widehat{A}$`)` → `STHGRec`
→ `[ ablations ]  [ honest / ungrounded + faithfulness ]`

Two isolated Python envs (recommender: RecBole + PyTorch; sentiment: PyABSA). They never share a process; they meet only through files in `data/`. Every run is driven by one `configs/*.yaml` at `seed 2020`.

3 Baselines

Popularity, BPR, NGCF, LightGCN — all through vanilla RecBole 1.2.1. Pairwise BPR objective, uniform 1 negative / positive, full-ranking evaluation. Metrics Recall@K and NDCG@K for $K \in \{10, 20\}$.

4 Sentiment (ABSA)

PyABSA `FAST_LCF_ATEPC` (multilingual) over each review → open-vocabulary pairs (aspect, p) with polarity $p \in \{\text{Positive}, \text{Neutral}, \text{Negative}\}$. A separate normalizer maps raw aspects onto a fixed **12-term canonical vocabulary** (the 4 PRD-locked aspects *gameplay, graphics, price, difficulty* first, then 8 domain aspects: story, controls, sound, multiplayer, performance, value, . . .). Matching is word-boundary / word-start-prefix regex, case-insensitive, **first canonical in insertion order wins**; unmatched terms are dropped and reported for curation. Output: `absa_normalized.csv` with `user_id, item_id, aspect, sentiment`.

5 Trust signals (per user, Option 4A)

Computed over each user's full review history (a slow-moving reputation *prior*, not a per-interaction label). With σ_u the std. of user u's star ratings:

$$\text{verified}_u = \text{mean}(\text{verified_purchase}), \qquad \text{consistency}_u = \frac{1}{1+\sigma_u},$$

$$\text{helpful}_u = \frac{\text{mean}_u(\text{helpful_vote})}{\max_v \text{mean}_v(\text{helpful_vote})}, \qquad \boxed{\text{trust}_u = \tfrac{1}{3}\big(\text{verified}_u + \text{helpful}_u + \text{consistency}_u\big) \in [0,1]}$$

6 Graph → tensors

Node index space (one global table, block-offset; sorted raw IDs ⇒ deterministic, seed-independent; persisted to `sthgrec_id_maps.json`):

$$[0,\ n_u) \text{ users} \qquad [n_u,\ n_u{+}n_i) \text{ items} \qquad [n_u{+}n_i,\ n_u{+}n_i{+}n_a) \text{ aspects.}$$

Three relations, each a weighted edge list, de-duplicated / aggregated:

- *interacts* $(u{-}i)$: binary, one edge per pair. Weight $w_{ui} = \text{trust}_u$ if `use_trust_edges` else 1 (missing trust → 1).
- *sentiment* $(u{-}i)$: signed polarity $s(p) \in \{+1, 0, -1\}$; $w_{ui} = \text{clip}\big(\text{mean}_{\text{aspects}}\, s,\ -1,\ +1\big)$.
- *has_aspect* $(i{-}a)$: $w_{ia} = \text{mean}_{\text{rows}}\, s(p)$.

Symmetric normalization per relation (LightGCN style, sign-aware):

$$\widehat{A} = D^{-1/2}\,(A + A^\top)\,D^{-1/2}, \qquad D_{ii} = \sum_j \big|\,(A + A^\top)_{ij}\,\big|.$$

Degree uses $|w|$ so negative sentiment does not cancel the normalizer; $\widehat{A}$ keeps the sign. Neutral ($s = 0$) edges are present but carry no message.

7 Model — ST-HGRec

One embedding table $E \in \mathbb{R}^{(n_u+n_i+n_a)\times d}$, $d = 64$, Xavier init. **No feature transforms, no nonlinearities, no attention, no message dropout** — LightGCN discipline; sentiment and trust are purely structure and weights.

$L = 3$ **layer heterogeneous propagation**, then layer-mean:

$$E^{(0)} = E, \qquad E^{(l)} = \widehat{A}_{\text{int}} E^{(l-1)} \ +\ \lambda_s\, \widehat{A}_{\text{sent}} E^{(l-1)} \ +\ \lambda_a\, \widehat{A}_{\text{asp}} E^{(l-1)},$$

$$E_{\text{final}} = \frac{1}{L+1} \sum_{l=0}^{L} E^{(l)}, \qquad \hat{y}_{ui} = E_{\text{final}}[u]^\top E_{\text{final}}[i].$$

The sentiment / has_aspect terms are dropped entirely when `use_sentiment_edges` is false. $\lambda_s = \lambda_a = \lambda_t = 1$ are **fixed** (config-exposed, not learned). Aspect nodes have no direct user edge — their signal reaches users only via item → aspect → item paths, so $L \geq 2$ matters.

Loss (BPR + L2, identical objective to the RecBole baselines):

$$\mathcal{L} = \sum_{(u,i^+,i^-)} -\ln \sigma\big(\hat{y}_{ui^+} - \hat{y}_{ui^-}\big) \ +\ \lambda_{\text{reg}} \big\| E^{(0)}_{u,\,i^+,\,i^-} \big\|_2^2.$$

$\lambda_{\text{reg}} = 10^{-4}$, Adam, lr 10^{-3}, batch 2048, ≤300 epochs, early stop on NDCG@10, 1 uniform negative / positive.

Harness. Vanilla RecBole `Config/Trainer/evaluate` (Option B). **Propagation** is hand-written `torch.sparse.mm` over the three coalesced COO adjacencies (Option C) — `torch_geometric` is pinned but imported nowhere. A RecBole-id ↔ global-node translation layer maps RecBole's internal ids to our sorted index; unseen valid/test ids route to a zero PAD slot (matches baseline cold-start).

8 Ablations (switch-off matrix)

variant	`use_sentiment`	`use_trust`	purpose
full	true	true	headline ST-HGRec number
−sentiment	false	true	isolate sentiment value
−trust	true	false	isolate trust value
−both	false	false	≈ LightGCN (interacts-only) — sanity anchor

Each flag flip *omits the whole relation term* from the propagation sum (the spmm is skipped, not zeroed) — exact and cheap. `use_trust_edges=false` reverts every interaction weight to 1, so −both must reproduce the LightGCN baseline within tolerance; a closed-form 5-node parity test asserts exactly this.

9 Explanations and faithfulness

Honest (`explain/honest.py`): `extract_facts` reads only the model's hooks — top-K aspects by |contribution|, the verbatim supporting sentiment rows, and the trust context. A deterministic template renders 2–4 sentences citing *only* those aspects. An optional injectable `rephrase_fn` can hand the facts block to an LLM at study time — **no LLM SDK is imported anywhere in `src/`**.

Ungrounded (`explain/ungrounded.py`): fixed persuasive templates keyed by item identity, referencing no graph fact. **Quarantined** — an AST test forbids any `src/models`, `src/eval`, `src/graph`, `src/data`, or `src/sentiment` module from importing it.

Remove-a-reason faithfulness (`explain/faithfulness.py`): for each cited aspect a of item i, delete the (i, a) sentiment rows, rebuild $\widehat{A}$, and re-propagate the model's *frozen* embeddings (only structure changes). Record $|\Delta\text{score}| = |\text{score}' - \text{score}|$ and whether a's `has_aspect` contribution collapses. Then

$$\text{faithfulness} = \frac{\#\{\text{reasons with } |\Delta\text{score}| > \varepsilon \text{ or contribution collapse}\}}{\#\text{cited reasons}}, \qquad \text{grounding} = \frac{|\text{cited} \cap \text{real}|}{|\text{cited}|}.$$

On a connected graph an honest explanation scores faithfulness = 1.0; an ungrounded blurb cites no real aspect, so its grounding = 0.

10 Results (seed 2020, official split)

model	Recall@10	Recall@20	NDCG@10	NDCG@20
Pop	0.0050	0.0071	0.0032	0.0039
BPR	0.0073	0.0111	0.0049	0.0060
NGCF	0.0075	0.0135	0.0049	0.0067
LightGCN	0.0079	0.0131	0.0047	0.0063
STHGRec (no-sentiment sanity)	**0.0094**	**0.0142**	**0.0056**	**0.0071**

Ordering is sane (LightGCN $\geq$ NGCF $\geq$ BPR $\geq$ Pop). The whole benchmark runs $\sim$10$\times$ lower than pre-correction random-resplit numbers — expected, the temporal split is harder. The STHGRec row is a **sanity run with the sentiment layer empty** (the full ABSA output is still being generated); it is not the headline number, and the −sentiment / −trust / −both ablation rows are still to come.

11 CPU vs GPU

- **ABSA is the long pole.** `FAST_LCF_ATEPC` runs $\approx$9 reviews/s on this Mac (Apple MPS-assisted); 748K reviews $\approx$ 16–20 h wall-clock, checkpointed and resumable. A CUDA GPU is roughly 10–50$\times$ faster ($\sim$1–2 h). Run the full ABSA pass on a CUDA box.
- **ST-HGRec is forced CPU.** Apple MPS has no sparse-tensor kernels, so `torch.sparse.mm` propagation cannot run there; `configs/sthgrec.yaml` pins `device: cpu`. A full CPU training run is a few hours; a CUDA GPU would cut it to minutes, but sparse `spmm` determinism must be re-verified there before reporting a headline number (CPU `torch.sparse.mm` is deterministic and passes the seed-stability bar today).
- **Not memory-bound.** Embedding table $\approx$15 MB; all sparse adjacencies <1 GB. The constraint is compute / kernel availability, not RAM — a modest CUDA card suffices.
- **Baselines** (RecBole LightGCN/NGCF): hours on CPU, minutes on GPU; numbers are split-deterministic either way.

12 Reproducibility spine

Two pinned requirements files; `seed 2020` for RecBole, the negative sampler, and NumPy; sorted ID maps remove dict-iteration nondeterminism; the ID-map JSON is committed and asserted-equal on load; every metric row traces to a `configs/*.yaml`. Heavy regenerable artifacts (`*.gpickle`, processed data, raw parquet) are git-ignored — only code, configs, and small result tables/figures are tracked.